AN EXACT SEQUENCE COMPARING LOCAL COHOMOLOGY ALONG A SPECIALIZATION-STABLE SUBSET WITH ITS NONMINIMAL POINTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Local cohomology with support in a specialization-stable subset, and its description as a filtered colimit over closed subsets. The author's lemma labelled lemma-adjoint-ext (source lines 996-1004), delivered as the context span of this card. Identifies the colimit term in the displayed exact sequence.
- The degeneration of the local cohomology spectral sequence of another chapter to short exact sequences. Invoked inside the counted proof at source line 1236 and attributed to the cited chapter's lemma (a cross-chapter pointer). Supplies the short exact sequences that the proof assembles into the displayed sequence.

Source reference. Original reference dualizing-lemma-local-cohomology-ss: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 428–428. Navigation pointer only: the counted text refers to the Stacks item dualizing-lemma-local-cohomology-ss, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. Let $T' \subset T$ be the set of nonminimal primes in T. Then T' is a subset of $\mathrm{Spec}(A)$ stable under specialization and for every A-module M there is an exact sequence*

$$0 \to \mathrm{colim}_{Z,f} H^1_f(H^{i-1}_Z(M)) \to H^i_{T'}(M) \to H^i_T(M) \to \bigoplus\nolimits_{\mathfrak{p} \in T \setminus T'} H^i_{\mathfrak{p}A_{\mathfrak{p}}}(M_{\mathfrak{p}})$$

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where the colimit is over closed subsets $Z \subset T$ and $f \in A$ with $V(f) \cap Z \subset T'$.

Lemma 0.2. *Let A be a Noetherian ring. Let $T \subset \mathrm{Spec}(A)$ be a subset stable under specialization. For any object K of $D(A)$ we have*

$$H^i(RH^0_T(K)) = \mathrm{colim}_{Z \subset T \text{ closed}} H^i_Z(K)$$

Proof. For every Z and f the spectral sequence of Dualizing Complexes, Lemma dualizing-lemma-local-cohomology-ss degenerates to give short exact sequences

$$0 \to H^1_f(H^{i-1}_Z(M)) \to H^i_{Z \cap V(f)}(M) \to H^0_f(H^i_Z(M)) \to 0$$

We will use this without further mention below.

Let $\xi \in H^i_T(M)$ map to zero in the direct sum. Then we first write ξ as the image of some $\xi' \in H^i_Z(M)$ for some closed subset $Z \subset T$, see Lemma 0.2. Then ξ' maps to zero in $H^i_{\mathfrak{p}A_\mathfrak{p}}(M_\mathfrak{p})$ for every $\mathfrak{p} \in Z$, $\mathfrak{p} \notin T'$. Since there are finitely many of these primes, we may choose $f \in A$ not contained in any of these such that f annihilates ξ'. Then ξ' is the image of some $\xi'' \in H^i_{Z'}(M)$ where $Z' = Z \cap V(f)$. By our choice of f we have $Z' \subset T'$ and we get exactness at the penultimate spot.

Let $\xi \in H^i_{T'}(M)$ map to zero in $H^i_T(M)$. Choose closed subsets $Z' \subset Z$ with $Z' \subset T'$ and $Z \subset T$ such that ξ comes from $\xi' \in H^i_{Z'}(M)$ and maps to zero in $H^i_Z(M)$. Then we can find $f \in A$ with $V(f) \cap Z = Z'$ and we conclude. $\square$